

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



448/1

Electricity

Paper 1

Time: 2½ Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- a) Write your **Name**, **Index number** and **Class** in spaces provided
- b) Candidates should have the following for this examination
Drawing instruments
Scientific calculator
- c) This paper consists of two sections: **A** and **B**.
- d) Attempt **ALL** questions in **section A** and any **FOUR** in **section B**.
- e) All dimensions are in **millimeters** unless stated other wise

For Examiner's use only

Section	Questions	Maximum Score	Candidate's Score
A	1-10	48	
B	11	13	
	12	13	
	13	13	
	14	13	
	15	13	

SECTION A (48 marks)

*Attempt **all** questions in this section in the spaces provided*

1. a) State the most immediate action to be taken when rescuing a worker found unconscious and still in contact with a live wire (1 mark)

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- b) List two safety precautions taken while drilling through a piece of sheet metal (2 marks)

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- c) State the role of Directorate of Industrial Training (1 mark)

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- d) Outline four major items that must constitute a business plan (2 marks)

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2. a) The range of a moving iron instrument of resistance 8 ohms is extended by using a shunt resistor of 0.04 ohm. If the full scale deflection current is 10mA, determine the maximum current that the modified instrument can measure (2 marks)

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b) State two advantages and two disadvantages of the moving iron instrument over moving coil instrument (2 marks)

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3. a) List two power company equipment and two consumer’s equipment found at a domestic consumer’s intake point. (2 marks)

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b) State one advantage and one disadvantage of fluorescent lamps over filament lamps (2 marks)

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4. State two reasons why the rotating field alternator is preferred to the rotating armature type (2 marks)

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5. a) After installing new cells in a 2 cell torch it failed to light when switched on. State two possible causes for the failure (2 marks)

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b) Three capacitors are connected as shown in **Figure 1**

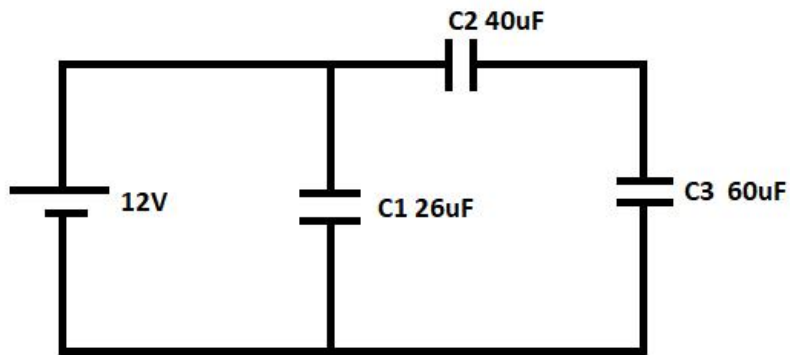


Figure 1

Calculate:

i. The equivalent capacitance

ii. Charge stored in the 26 μ F capacitor **(4 marks)**

6. Sketch and label the schematic diagram of a dc shunt motor connected to a supply **(4 marks)**

7. a) Name the materials used to make the following parts of ac machine in **Table 1** and in each case state one reason why the material is used (3 marks)

Table 1

Part	Material	Reason
Brushes		
Slip rings		

b) State three factors to consider when choosing motor for a particular installation (3 marks)

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8. a) List four layout tools used in metal fabrication (2 marks)

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b) Describe three characteristics of magnetic lines of force (3 marks)

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9. a) Name two types of inductor cores (1 mark)

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b) State two factors that determine inductance of a coil (2 marks)

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10. a) Determine the resistance of each of the following carbon resistors if their colour code is:

i. Blue Orange Brown Silver

ii. Grey Red Yellow (2 marks)

b) Figure 2 shows a resistor network

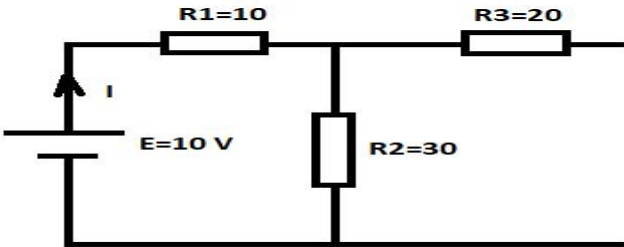


Figure 2

Find value of current I (2 marks)

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c) Name the logic gate shown in figure 3 below

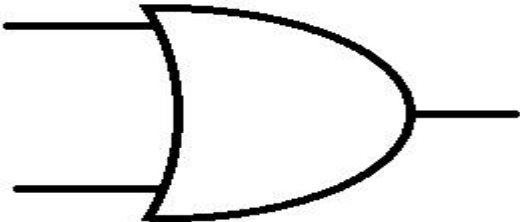


Figure 3

Name

d) Complete the truth table of the AND gate shown in **Table 2** (2 marks)

INPUT 1	INPUT 2	OUTPUT
0	0	
0	1	
1	0	
1	1	

Table 2

SECTION B (52 marks)

Answer any four questions from this section in the spaces provided

11. Figure 4 below shows a block drawn in isometric projection

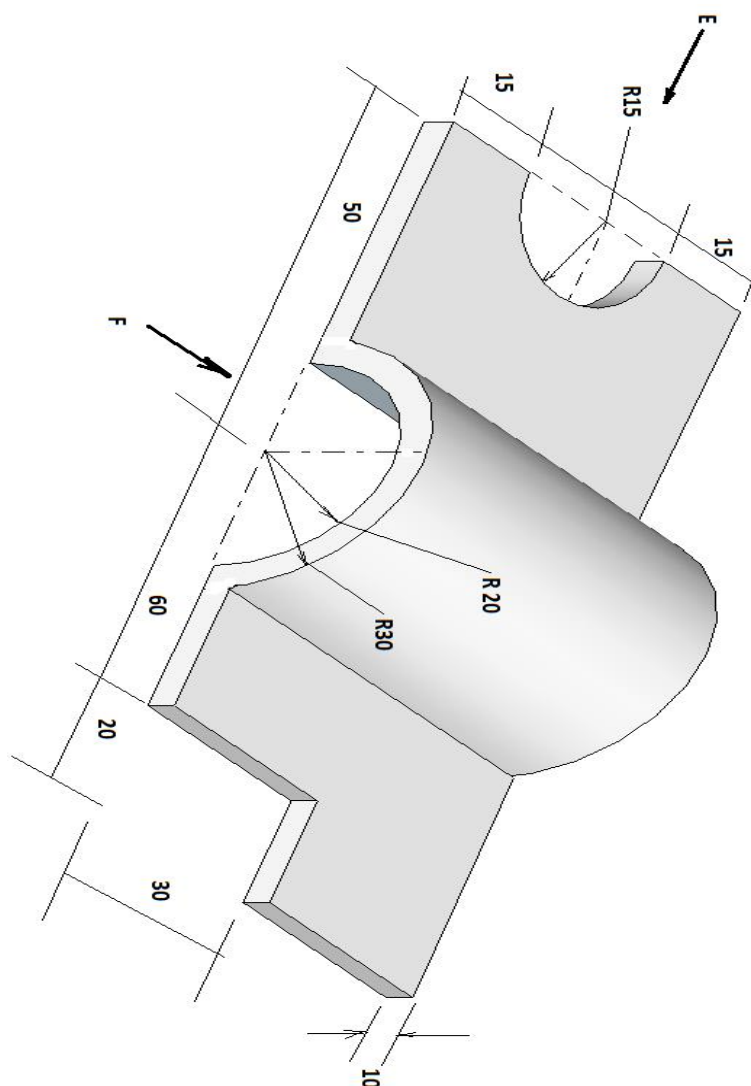


Figure 4

Using first angle orthographic projection, draw the following views full size

- a) Front elevation in the direction of arrow F
 - b) End elevation in the direction of arrow E
 - c) Plan
- (13 marks)**

DRAW HERE

12. a) State three advantages of toroidal type transformer over shell type transformer

(3 marks)

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b) i) Outline three assumptions made in order to consider a transformer as an ideal machine

(3 marks)

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ii) A5000/500V, 10KVA ideal single phase transformer has 40 turns on the secondary.

Calculate:

I. primary turns

II. primary full load current

III. secondary full load current

(7 marks)

13. a) State the phase relationship between current and voltage in circuits that are purely:

i. Resistive

ii. Inductive

(2 marks)

b) Figure 5 shows an RLC circuit

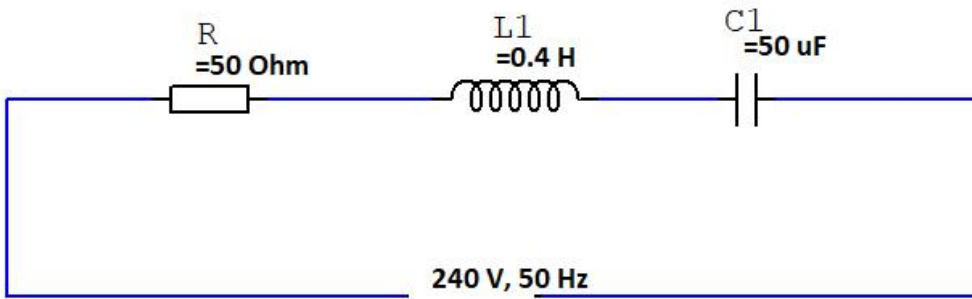


Figure 5

Calculate the:

i. Inductive reactance

ii. Capacitive reactance

iii. Circuit impedance

iv. Circuit current

v. Power dissipated in the circuit

(11 marks)

14. a) List three metal parts that are exempted from earthing in a domestic installation

(3 marks)

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b) Give three reasons why a verification of polarity test is carried out in a completed domestic installation

(3 marks)

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- c) Draw a line diagram of a national grid system showing typical voltages at each stage
(7 marks)

15. a) Draw a labelled circuit diagram of a PNP transistor amplifier in common base configurations
(3 marks)

b) **Table 3** below shows a bill of materials used to construct a stabilized dc supply

No	Description of material	Quantity
1	Double pole switch	1
2	Double wound transformer	1
3	Rectifier diodes	4
4	Electrolytic capacitor	1
5	Carbon resistors	1
6	Zener diode	1
7	Connecting wires	Several

Table 3

Draw a circuit diagram of the power supply

(10 marks)